

3 A Velocity of motorcyclist relative to car, $\mathbf{V}_R = \mathbf{V}_M - \mathbf{V}_C$

4 C Separation between P and Q is maximum when P is just hitting the ground.

Time for P to reach the ground is given by:

$$s = ut + \frac{1}{2}gt^2$$

$$= \frac{1}{2}gt^2$$

$$80 = 5t^2$$

$$t = 4 \text{ s}$$

Position of Q when P is hitting the ground:

$$s = u(t-1) + \frac{1}{2}g(t-1)^2$$

$$= \frac{1}{2}g(t-1)^2$$

$$= 5 \times (4-1)^2$$

$$= 45 \text{ m}$$

Hence, max. separation between P and $Q = 80 - 45 = 35 \text{ m}$

5. D. As the object is in static equilibrium, the resultant force of the three external forces acting on the